

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

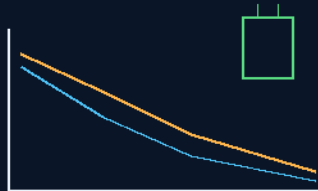
PHASE 236

THE DEGRADATION CATALOGUE

TIME TIMES TEMPERATURE IS THE DOSE

curves, not numbers

heat, light, oxygen — the three axes



honey's 35 — label is not plate — GMO by merit

ARRHENIUS RUNS IT · CONSERVATIVE UNTIL MEASURED

THE BAG LISTS THE HARVEST, NOT THE PLATE

Preservation science — v1.0 in force

GP-IM-236

Revision v1.0 — 24 August 2026

237 of 290

VETTED BATCH 33 — PASS · CONTINUING MEASUREMENT

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 236

PHASE 236: THE DEGRADATION CATALOGUE — STATUS: CURRENT — PASS; THE CATALOGUE ITSELF IS A CONTINUING MEASUREMENT PROGRAMME (VET BATCH 33). A standing ship document in its own right: the catalogue of how everything dies slowly. The doctrine: time multiplied by temperature is the dose — every food, vitamin, enzyme, substrate and feedstock gets its own degradation curve, with heat, light and oxygen as separate axes and time as the exposure multiplier. It began with a small man and a winter: the honey producer whose decanting line stalled, and the Electrotherm jacket job that taught the Author the industry's own spec — roughly 35°C jacket limit, 25-30°C product target, longer exposure at lower temperature — because honey's antibacterial engine is a bee-added enzyme, and the enzyme dies warm. LABEL ≠ PLATE seats with it: the bag lists what went in at freezing, not what lands on the plate. The enzyme gap is corrected gently — glucose oxidase already works commercially; scattered research, not absent. GMO BY MERIT closes it: Earth's politics stay on Earth; off-world, an engineered crop rides by choice where it earns the ride. And 235's oxygen-free gas loop connects: atmosphere composition is part of preservation, not an unrelated subsystem. An unmeasured curve stays conservative until measured — the catalogue's own law.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-236, v1.0 — 24 August 2026. The two hundred and thirty-seventh manual of the program — 237 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the catalogue law as seated (contents line, the doctrine — dose, the three axes, label-not-plate, substrates, the enzyme gap, GMO by merit) — §2 the physics, graded — §3 the system in the catalogue — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 236: **The Degradation Catalogue (Time × Temperature Is the Dose; Honey's 35; Label ≠ Plate; Heat, Light, Oxygen; GMO by Merit)**'

THE CORE OBJECTIVE (verbatim): 'Core Objective: a standing ship document in its own right — the degradation catalogue: every food, vitamin, enzyme, substrate and feedstock with its degradation CURVE (never a bare number) and its three-axis budget of heat, light and oxygen, with time as the multiplier — maintained as living law by the officer pair, binding on every process from the galley to the mold bank.'

Catalogue Baseline (graded this sitting): the honey spec is the industry's own — the antibacterial engine is a bee-added enzyme (glucose oxidase, manufacturing peroxide on dilution; manuka's medical grade adds a second, non-peroxide pathway), heat-labile above the 30-35 C band, so decanting holds ~35 C at the jacket and 25-30 C at the product; the industry even tracks a marker molecule for heat damage in every export standard. 'Longer exposures or higher temps shorter exposures' is the Arrhenius law of thermal death — the same D-value/z-value mathematics that runs every pasteurizer. Provenance preserved: Electrotherm, the heating-jacket line, IBC base units for honey and paints and glues, the smaller local guy whose line stalled every winter.

Live Instruction Confirmed: Yes — 3 August 2026, morning sitting.

Live instruction, 3 August 2026.

"Weirdly had to learn everything about honey for a client at electrotherm, a heating jacket for line production, larger clients bought heating base units and jackets for IBC containers for honey and paints and glues etc for decanting. / 35 c max so 25-30 max at product for decanting, it was a smaller local guy who had trouble with his line production in winter. heating a viscosity I already understood well, but honey has several grades, basic shop, high grade shop and medical, Honey has enzymes that break down over 30-35c, longer exposures or higher temps shorter exposures. the enzyme is what makes it antibacterial."

"we need to note that for all food productions growth etc, as vitamins and other food nutritional components are equally degraded by temperature. a bag of frozen vegetables nutritional listing was at fresh before frozen and before cooking."

"notably absent is any research into finding a way to add that enzyme into other food products by DNA or other methodologies. / whilst frowned on here, in space GMO may be our savior. / certainly understanding degradation points in all food and substrate materials is a massive task and required catalogue for the ship in its own right."

THE DOCTRINE — AS GIVEN, STRUCTURE READ (KIMI) (VERBATIM)

- **Degradation is a dose — the catalogue holds CURVES, not numbers.** Time × temperature is the unit, per item, per axis. There is no safe temperature, only a budget: hold it shorter or hold it cooler, the curve is the law. Every entry carries its time-temperature curve, its light sensitivity, its oxygen sensitivity — the three axes, time the multiplier.
- **The three axes — heat, light, oxygen.** Vitamins die by all three: C is the fragile one (heat, oxygen, water), the Bs leach and break down, riboflavin dies by light through a clear bottle. The catalogue's third axis is already fought by the ship: 235's oxygen-free scrubber-gas loop IS nutrition-preservation technology — the process design and the nutrient law are one document now.
- **LABEL ≠ PLATE — seated, with the audit's honest wrinkle.** The Author's exhibit: a bag of frozen vegetables lists its nutrition at fresh, before freezing and before cooking — the number on the bag is the value before your kitchen got to it. The wrinkle: flash-frozen at the field can BEAT 'fresh' that spent a week on a truck; the big thieves are boiling water (C and B leach out), storage time, and reheating. The catalogue records the whole path — harvest to plate — because the label never does.
- **Substrates ride the same law.** Polymers have processing windows and thermal histories (every reheat spends from the budget; some resins hydrolyze wet, some off-gas hot); feedstocks per 221's shelf-life law (bio for the consumable, made to order); the mold bank (234) consults the catalogue before every run — the material's remaining budget decides the process, not habit.
- **The enzyme gap — corrected gently: scattered, not absent.** Glucose oxidase already works as a commercial food preservative (oxygen-scavenger class), and antimicrobial enzymes have been expressed in crops — lysozyme- and lactoferrin-rice is real and licensed. The Author's direction is right; the research exists in pockets nobody connected. The catalogue is where the pockets meet.
- **GMO BY MERIT — the clause, seated.** Earth's politics stay on Earth. Off-world, an engineered crop rides by CHOICE where it earns it — biofortified yields, oxygen-scavenging enzymes, closed-loop traits — examined under the gnome law (234) like every other entry in the bank: nothing rides by default, everything rides by justification. The frown is not a law; the budget is.

PHYSICS NOTE — THE AUDIT (KIMI: ONE EQUATION RUNS THE WHOLE CATALOGUE) (VERBATIM)

Thermal degradation follows Arrhenius kinetics — the same exponential that sets pasteurization's D-values (time at temperature for a ten-fold reduction) and z-values (degrees per ten-fold rate change). This is why the catalogue can be a CURVE: one equation per entry, two constants, and the whole time-temperature trade is computable for any process — the decanting jacket, the freeze dryer's heat stage (235), the polymer mold (234), the galley's pot. THREE BENCH NOTES. ONE — the catalogue binds the label law: ship-grown and ship-processed foods are logged at plate value, not seed value — the galley

answers to the curve. TWO — water activity joins the axes for storage (honey's own first defense is being too dry to spoil — drying IS preservation, the oldest entry in the catalogue). THREE — the catalogue is the officer pair's living document (223): every new food, substrate, or feedstock enters with its curve measured or its curve flagged as unmeasured — unmeasured means conservative process until proven. ('decanting', '35 c max so 25-30 max at product' kept verbatim; readings standard.)

ORIGIN OF THOUGHT (VERBATIM)

It began, as the best ship laws do, with a small man and a winter. A local honey producer whose decanting line stalled when the temperature dropped; a heating jacket; a number learned the hard way — thirty-five at the jacket, thirty at the product, because past that line the thing that makes honey medicine gives up and dies. The Author filed that number for forty years and this morning generalized it: everything good is like honey. Vitamins, enzymes, feedstocks, polymers — all of them dying quietly by degrees and hours, and all of them listed on some label as if they weren't. So the ship gets a catalogue — not of foods but of fragilities — every entry a curve, every process answerable to it, the truth told at the plate and not at the seed. And because a catalogue of what dies is also a map of what to build, the GMO clause went in beside it: out here, an engineered crop is a tool to be examined, not a politics to be inherited. The honey man taught the ship its law: heat is a budget, and everything spends it.

Why it matters, in plain English: Honey stays antibacterial only because of an enzyme that dies if it gets too warm — so honey is warmed gently, never above about 35 degrees, and less for longer. The same is true of everything we eat: vitamins and nutrients are destroyed by heat, light and air, a little at a time — and the number on the packet is measured before any of that happens. This phase creates a ship's catalogue: for every food and material, a chart of exactly how much heat, light and air it can take before it dies — so cooking, drying, storing and molding never silently destroy the very thing they're trying to preserve. It also says plainly: in space, genetically improved crops are allowed if they prove themselves — because feeding a crew forever may depend on them.

Build the Degradation Catalogue: the ledger of how every material and system ages in service — everything degrades, and the catalogue records how fast, so replacement is scheduled, not surprised. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

DEGRADATION IS A DOSE, AFFIRMED AS THE CATALOGUE'S UNIT: time × temperature per item per axis — the catalogue holds CURVES, not numbers. There is no universal safe temperature; different materials carry different mechanisms, and the master requires the unmeasured curve to remain conservative until measured.

ONE EQUATION RUNS THE CATALOGUE, AFFIRMED: thermal degradation follows Arrhenius kinetics — the same exponential behind pasteurization's D-values and z-values. The treatment is appropriate and the file applies it honestly: rate doubles and halves with temperature steps; exposure multiplies.

THE THREE AXES ARE THE RIGHT ANATOMY: heat, light, oxygen — vitamin C is the fragile one (heat, oxygen, water), the Bs leach, the fats oxidise, the enzymes denature. The catalogue's structure matches the chemistry's structure.

THE HONEY SPEC IS THE INDUSTRY'S OWN, SEATED WITH PROVENANCE: the Electrotherm jacket job — the decanting line that stalled in winter — taught the 35°C jacket limit and the 25-30°C product target; the antibacterial engine is a bee-added enzyme with a thermal budget. Testimony class, and it checks.

LABEL ≠ PLATE AND GMO BY MERIT SEAT AS POLICY LAW: the label lists the harvest, not the plate — the honest wrinkle the audit adds is that the catalogue exists precisely to close that gap with data. GMO by merit is a design rule, not a physics claim — off-world the engineered crop rides where it earns the ride; the vet returned no physics objection.

§3 — THE SYSTEM IN THE CATALOGUE

- **The dose law:** time × temperature per item per axis — curves, not numbers.
- **The three axes:** heat, light, oxygen — each material dies its own way on each.
- **Honey's 35:** the jacket limit and product target — the enzyme budget written by industry.
- **LABEL ≠ PLATE:** the bag lists the harvest; the catalogue measures the gap.
- **The enzyme gap:** scattered, not absent — glucose oxidase already works commercially.
- **GMO BY MERIT:** Earth's politics stay on Earth; the crop rides where it earns the ride.
- **The conservative law:** unmeasured curve = conservative assumption until measured.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The Electrotherm testimony continues the file's provenance doctrine — the Author's industrial past keeps becoming ship law (TRS dried the polymer; Electrotherm heated the honey).
- The catalogue is 235's software: the vac pack and freeze dryer are the hardware that buys time; the catalogue prices what the time costs.
- The 221 loop connects — feedstock is perishable inventory; the catalogue is the countdown's ledger; the gas loop's atmosphere is a preservation axis, not a separate subsystem.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 33 (17 August 2026 — phases 236-240): 'Phase 236 — The Degradation Catalogue — PASS — with the catalogue itself being a continuing measurement programme. This phase is stronger than simply saying "food degrades with heat." The design makes degradation a stored process variable: each food, vitamin, enzyme, substrate and feedstock gets its own degradation curve, with heat, light and oxygen as separate axes and time acting as the exposure multiplier. The honey example is grounded in the author's actual industrial experience: the seated instruction records a roughly 35 °C jacket limit and 25–30 °C product target for decanting... The underlying thermal-degradation treatment is appropriate: Arrhenius-type kinetics are used... with D/z values where applicable... The catalogue does not assume one universal "safe temperature." That is exactly right... the master explicitly requires an unmeasured curve to remain conservative until measured... The GMO provision is a policy/design rule, not a physics claim, so it doesn't receive a physics objection. 236 → PASS.'

§6 — BUILD SEQUENCE

- 1. Open the catalogue: the standing document, curve format per item per axis, the conservative-default rule.
- 2. Seat the foundation curves: honey's jacket spec, the vitamin fragility table, the substrate thermal histories.
- 3. Wire the store: hold temperatures and gas loops driven by the catalogue, not by habit.
- 4. Run the measurement programme: the unmeasured curves benched in order of consumption volume.
- 5. Audit the labels: label-versus-plate sampling against the curves — the gap made data.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 235 (the vac pack and freeze dryer):** the hardware that buys the time the catalogue prices.
- **Phase 221 (the feedstock question):** the countdown stock the catalogue inventories.
- **Phase 204 (the must-haves):** the medical stores' own degradation law — iodine and antibiotics keep their budgets too.

§8 — DO-NOT-BUILD

- Do NOT quote one safe temperature — the catalogue holds curves; the universal number is the error.

- Do NOT trust the label at the plate — LABEL ≠ PLATE is law; measure the gap.
- Do NOT heat honey past its budget — the enzyme is the product; 35 is the jacket ceiling.
- Do NOT let an unmeasured curve ride optimistic — conservative until measured is the catalogue's own law.
- Do NOT import Earth's GMO politics — by merit, where it earns the ride; policy seated as policy.

§9 — OWED

OWED: the continuing measurement programme itself — the curve bench for the top-volume items, the label-versus-plate audit data, the enzyme-addition literature watch (scattered research tracked), and the store-temperature wiring verified against the catalogue.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-236, v1.0 — CURRENT — PASS; THE CATALOGUE ITSELF IS A CONTINUING MEASUREMENT PROGRAMME (VET BATCH 33), seated law, master v2.1 (numerical print order). The two hundred and thirty-seventh manual of the program — 237 of 290. Built 24 August 2026, seventeenth of the 20-manual 'ok ill do these 20, you do 20 more' shift (the author's order, 24 August 2026, seated in sitting 618). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607 and 618): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more'. The phase's own chain, all seated in the master: the contents line, the masthead trio and the bodies (as seated); the live instruction of 10 August 2026 (seated per sitting 146) — time × temperature as the dose; heat, light and oxygen as separate axes; the honey testimony: 'Weirdly had to learn everything about honey for a client at electrotherm', verbatim — the 35 °C jacket limit and the 25-30 °C product target; LABEL ≠ PLATE; the enzyme-gap correction (glucose oxidase scattered, not absent); GMO BY MERIT; the rename of 12 August 2026; the vet batch 33 grade (PASS — the catalogue itself a continuing measurement programme) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the first measured-curve batch, or his word.

END OF MANUAL — PHASE 236